This Bird's Wings Are Made for Sharp Turns

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The European nightjar is a bird that hunts insects at night. Every year, it flies thousands of miles between Europe and southern Africa.

You might think a bird that travels so far would have wings made for flying fast. But scientists at Lund University found something different.

They studied nightjars in a **wind tunnel**. The birds have broad wing tips. These help them make sharp turns while flying slowly to catch insects. But when the birds fly fast, their wings make **lift** less well. **Lift** is the force that helps keep a bird in the air.

Scientist Christoffer Johansson said the wings are "a **compromise**," a mix that balances different needs.

The team found a surprise, too. When the nightjar raises its wings, it still pushes itself forward. Scientists had mainly seen this in bats and insects.

This idea could help **engineers** build better flying robots and **drones**. Next, the team wants to know if other birds do the same thing.

Word Treasure

wind tunnel -- A tube where air is blown past something so scientists can study it.

lift -- The force that helps keep a bird in the air.

compromise -- A mix that balances different needs.

engineers -- People who design and build machines and structures.

drones -- Flying machines that have no pilot on board.

Background you need

Some birds fly very long distances each year to find food and warm weather.

Birds stay in the air because of a force called lift, and they move forward by pushing air with their wings.

Scientists study how animals fly to learn about nature and to build better machines.

Break it down

WHO: Scientists at Lund University, including Christoffer Johansson
WHAT: They found the European nightjar's wings are made for sharp turns, not just fast flying
WHERE: In a wind tunnel, studying a bird that flies between Europe and southern Africa
WHY: To learn how the bird's wings balance long trips with catching insects

Why it matters

Learning how birds fly could help engineers build better drones you might see one day.

Test what you remember

1. When does the European nightjar hunt insects?

- (A) Never
- (B) At night
- (C) Only in winter
- (D) At noon

2. Where did scientists study the birds?

- (A) In a wind tunnel
- (B) In a cave
- (C) In a swimming pool
- (D) On a mountain

3. What do the nightjar's broad wing tips help it do?

- (A) Build nests
- (B) Sing loudly
- (C) Swim fast
- (D) Make sharp turns

4. What is lift?

- (A) A kind of food
- (B) A bird's song
- (C) The force that helps keep a bird in the air
- (D) A type of feather

5. What surprise did scientists find?

- (A) The bird never sleeps
- (B) The bird can talk
- (C) The bird has four wings
- (D) The bird pushes forward even when raising its wings

6. What could this discovery help engineers build?

- (A) Bigger ships
- (B) Faster trains
- (C) Taller buildings
- (D) Better flying robots and drones

Answer key (try the quiz first!): 1: B 2: A 3: D 4: C 5: D 6: D

Questions to think about

1. Christoffer Johansson, scientist: Says the nightjar's wings are a compromise.

MY THOUGHTS (at least 20 words):